DAY_07

(Assignment)

1. Develop an Employee Management System that:

I. Uses pattern matching to process different employee objects.

2. Design and implement a Java-based Event Management System using the java.time.LocalDate class.

The system should allow users to:

- Create an event by entering its name, start date, and end date.
- Determine the duration of the event in days.
- Check whether an event is upcoming, ongoing, or completed based on the current date.
- Identify whether two events overlap with each other.
- Find the earliest and latest event from a collection of events.
- Calculate the number of days remaining until an upcoming event.
- Display all events scheduled within a specified date range.
- Handle invalid date ranges appropriately.

3. Online Banking System Develop a Java program for an Online Banking System that allows users to withdraw and transfer money.

The program should handle the following exceptions:

- Attempting to withdraw an amount greater than the available balance.
- Entering a negative or zero amount.
- Entering an invalid account number.
- Attempting to transfer money when the account has insufficient funds.

Create and use custom exceptions where appropriate.

4. Student Examination and Result System - Develop a Java application that accepts student marks for multiple subjects and calculates the total, average, and grade.

The program should handle:

- Invalid marks such as values below 0 or above 100.
- Non-numeric input entered by the user.
- Division-by-zero errors when calculating the average.
- Missing or invalid student information.

Implement appropriate built-in and user-defined exceptions and demonstrate the use of try, catch, finally, throw, and throws.

5. E-Commerce Shopping System

Develop a Java-based E-Commerce Shopping System where a customer can add products to a shopping cart and place an order.

The application should handle:

- Attempting to purchase a product that is out of stock.
- Entering an invalid product ID.
- Entering a negative or zero quantity.
- Insufficient account balance during payment.
- Attempting to place an empty order.

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